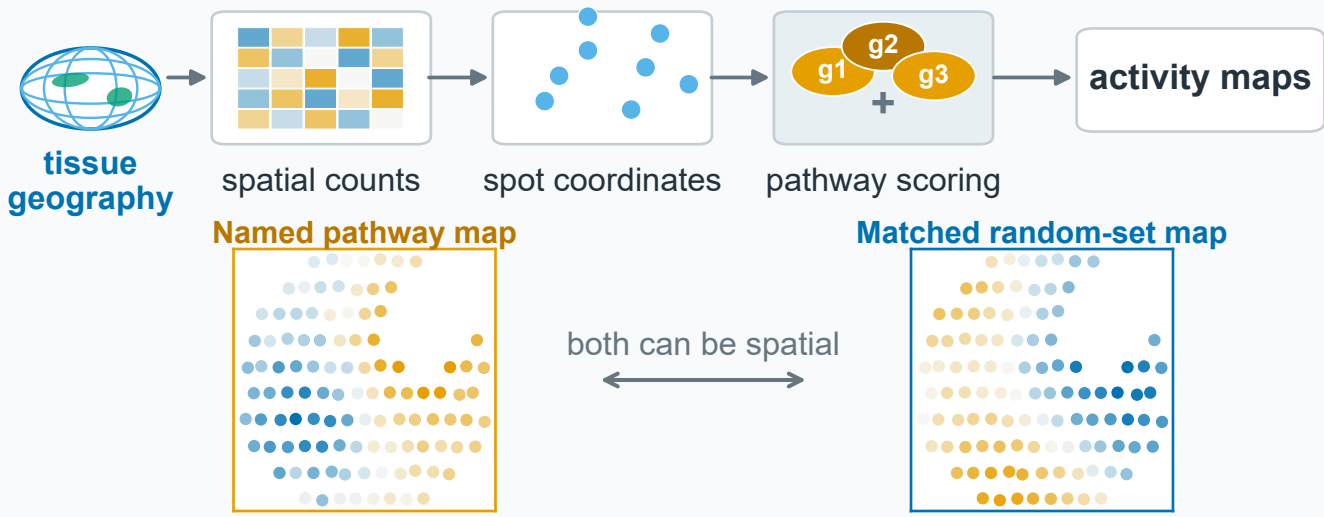


A Structured tissue makes many score maps spatial



Spatially patterned ≠ pathway-specific

B Absolute and competitive nulls ask different questions

Same observed score map and same spatial statistic
Moran / covariance / Gaussian process

ABSOLUTE TEST
permute spot labels within section

observed → permuted

Spatially associated?

Answer: any spatial association

COMPETITIVE MATCHED TEST
keep coordinates; change the gene set

observed → matched maps

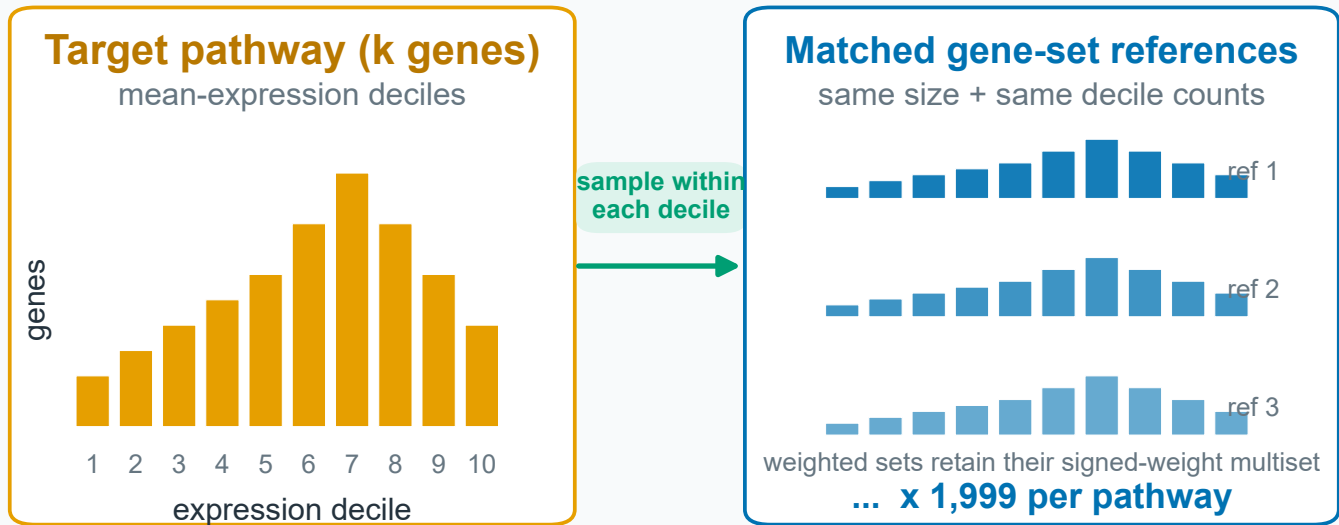
... x 1,999

Exceptional among matched sets?

Answer: pathway-specific exceptionality

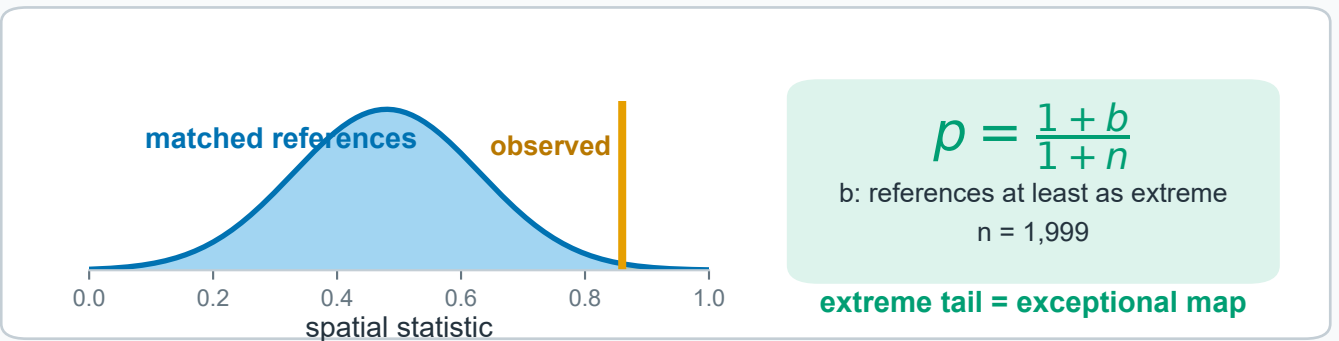
Different scientific questions - not an accuracy ranking

C Each pathway receives a matched reference class



same set size ✓
same decile profile ✓
original members excluded ✓
Not explicitly matched on within-set correlation or gene-network structure

D Exceptionality is judged within the matched distribution



Pt-1A pancreatic tumor: the reference changes the conclusion

Absolute endpoint

50/50 100/100
Hallmark random sets
spatial association is pervasive

change the scientific reference

Competitive matched endpoint

7/50
Hallmark pathways
only a subset is exceptional

Random sets are diagnostic comparators, not known negatives; counts reflect different nulls, not accuracy.